Evaluating Small Open LLMs for Medical Question Answering: A Practical Framework

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Abstract

Incorporating large language models (LLMs) in medical question answering flows demands more than high average accuracy: a model that returns substantively different answers each time it is queried is not a reliable medical tool, regardless of its mean quality score. Online health communities such as Reddit have become a primary source of medical information for millions of users, yet they remain highly susceptible to misinformation [23, 24]; deploying LLMs as assistants in these settings amplifies the need for output consistency alongside correctness. We present a practical, open-source evaluation framework for assessing small, locally-deployable open-weight LLMs on medical question answering, treating *reproducibility* as a first-class metric alongside lexical and semantic accuracy. Our pipeline computes ~~eight~~ a comprehensive suite of quality metrics, including BERTScore, ROUGE-L, and an LLM-as-judge rubric, together with ~~two~~ complementary lexical and semantic within-model reproducibility metrics derived from repeated inference (N=10 runs per question). Evaluating ~~three~~ four open-weight models (Llama 3.1 8B, Gemma 3 12B, Gemma 3 4B, and the clinically fine-tuned MedGemma 1.5 4B)~~on 50 MedQuAD questions (N=1,500 total responses) reveals that despite~~ , with the closed GPT-4.1-nano as an external reference, on two consumer-health benchmarks (MedQuAD and MedicationQA; 50 questions each, 10 runs per question, 5,000 responses in total), we find that low-temperature generation (T=0.2) ~~,~~ does not yield reproducible outputs: lexical self-agreement ~~across runs reaches at most 0.20, while 87–97% of all~~ is only 0.11–0.26 and 76–99% of outputs per model are unique~~, a safety gap that~~ . Yet the mean pairwise *semantic* self-similarity across runs is 0.94–0.97 for every model on both datasets, showing that this run-to-run variability is overwhelmingly paraphrastic rather than meaning-level—a distinction that purely lexical, single-pass benchmarks ~~entirely miss . We~~ miss entirely, and one that holds even for the closed reference model. With the same-size control included, we further find that ~~the clinically fine-tuned~~ model scale, not clinical fine-tuning, drives quality: MedGemma 1.5 4B ~~underperforms the larger~~ does not improve over its same-size general-purpose ~~models on both quality and reproducibility; however, because MedGemma is also the smallest model (4B vs. 8B and~~ sibling

Gemma 3 4B, while the larger Gemma 3 12B ~~), this comparison confounds domain fine-tuning with model scale, and a same-size comparison is needed to isolate the effect of clinical specialization~~leads the open models. All model-level estimates are reported with clustered bootstrap confidence intervals, and we quantify the reliability of the LLM judge itself (ICC(1) = 0.94 on MedQuAD, 0.69 on MedicationQA). We describe the methodology in sufficient detail for practitioners to replicate or extend the evaluation for their own model-selection workflows. All code, data processing pipelines, and experiments are released to ensure full reproducibility. The full implementation is available at https://github.com/aviad-buskila/llm_medical_reproducibility.

1 Introduction

Online medical communities such as Reddit have become a primary source of health information for millions of users, yet they remain highly susceptible to the rapid spread of misinformation and unverified advice [23, 24]. While large language models (LLMs) offer a promising opportunity to support question answering in these settings, their integration into community-driven platforms requires careful evaluation of reliability and safety. In this work, we explore the potential of small, locally deployable open LLMs as assistants in medical subreddits, with the goal of improving answer quality while mitigating the risks associated with inconsistent or misleading model outputs.

The availability of capable open-weight LLMs that can run entirely on local hardware has made clinical AI accessible to resource-constrained settings where patient data cannot leave institutional boundaries. Models such as Llama 3.1 [3], Gemma 3 [12], and domain-fine-tuned variants like MedGemma [8] can now be queried on a commodity workstation, lowering both the cost and the compliance burden of piloting LLM-based clinical decision support.

Yet the dominant evaluation paradigm for these models, benchmarking single-pass accuracy against a clinically validated ground-truth corpus, is fundamentally inadequate for medical usage. A drug-interaction query that returns a correct answer 70% of the time and a plausible but wrong answer the other 30% is not a useful medical tool; neither is one that returns a different correct answer on every invocation, because the instability itself erodes trust and auditability. In safety-critical applications, regulators and risk managers need to know not only *whether* a model is usually correct, but *how stable* those answers are.

Existing clinical NLP benchmarks, MedQA [4], PubMedQA [5], and others, evaluate single-pass accuracy and leave run-to-run variance unmeasured.

This paper makes the following contributions:

1. A **reproducibility-first evaluation protocol** that quantifies within-model run-to-run consistency via lexical

self-agreement ~~rate and response uniqueness rate~~ and response-uniqueness rates *and* a semantic self-similarity measure, alongside a standard accuracy suite, with all estimates reported as bootstrap confidence intervals.

2. A **modular, resumable, open-source pipeline** implementing the protocol with a simple three-stage CLI (`run` → `score` → `report`).

3. **Empirical results** on ~~three~~ five models—four small open-weight models ~~(N=1,500 responsesacross~~ spanning 4B–12B parameters plus a closed reference—across two consumer-health datasets (N=5,000 responses; 50 ~~MedQuAD questions~~ questions × 10 runs per model per dataset), demonstrating that reproducibility failures persist even in the near-deterministic temperature regime~~. We also observe that the clinically fine-tuned model underperforms larger general-purpose models, though this finding is confounded by a 2–3× parameter count difference~~, that run-to-run divergence is largely paraphrastic (high semantic self-similarity despite low lexical agreement), and that—using a same-size control—model scale rather than clinical fine-tuning drives quality.

2 Related Work

Self-consistency and output stability. Wang et al. [15] introduced self-consistency as a decoding strategy that samples multiple chain-of-thought paths and selects the most frequent answer, substantially improving reasoning accuracy. Chen et al. [16] extended this idea to free-form generation with universal self-consistency. Both lines of work use repeated sampling to *improve* output quality; our work repurposes the same repeated-sampling setup to *measure* output stability, turning the consistency signal itself into a first-class evaluation metric rather than a decoding technique.

LLM calibration and reliability. Kadavath et al. [17] showed that language models can partially assess the correctness of their own outputs, yet their confidence estimates remain imperfectly calibrated. Our finding that low-temperature generation (T=0.2) does not yield stable outputs complements this line of work: even when the sampling distribution is sharply peaked, the resulting text varies substantially across runs, suggesting that calibration and output determinism are distinct challenges.

Medical LLM evaluation. Singhal et al. [11] demonstrated that large language models encode clinical knowledge sufficient to approach expert-level performance on medical licensing exams, while Nori et al. [18] showed that generalist foundation models can match or exceed domain-specialized systems on medical benchmarks. However, both studies evaluate single-pass accuracy and do not report run-to-run variance, leaving the reproducibility dimension of clinical LLM behavior unexplored. Our work addresses this gap by treating within-model consistency as a co-equal evaluation axis alongside accuracy.

LLM-as-judge evaluation. Zheng et al. [14] established the LLM-as-judge paradigm for scalable open-ended evaluation, demonstrating strong agreement with human raters while also documenting systematic biases such as position and verbosity effects. We adopt a locally-hosted open-source judge to maintain full reproducibility and keep evaluation costs low, while acknowledging the inherent stochasticity of the judge itself as a limitation (Section 5).

3 Methodology

3.1 Datasets

We use **MedQuAD** [1], a collection of 16,412 medical question-answer pairs curated from fourteen NIH online health resources (NIHSeniorHealth, Genetics Home Reference, Cancer.gov, and others). Questions cover a wide range of clinical focus areas including symptoms, treatments, diagnoses, prevention, and prognosis. All content is in English and targeted at informed general audiences, making it a good proxy for consumer-facing clinical QA. Although MedQuAD is professionally curated rather than user-generated, its questions closely mirror the types of health inquiries commonly posted on platforms such as Reddit medical communities [24], covering symptoms, treatments, and prevention in lay-accessible language. We therefore use MedQuAD as a *proxy benchmark* for the Reddit-style consumer health questions that motivate this work: it provides clinically validated reference answers (which user-generated forum posts lack) while remaining representative of lay health information needs.

To test whether our findings generalize beyond a single corpus, we add a second benchmark, **MedicationQA** [2], a set of 689 consumer medication questions (covering dosing, interactions, administration, and side effects) paired with answers curated from authoritative drug-information sources (the MedInfo 2019 collection). MedicationQA is narrower and more drug-specific than MedQuAD and its reference answers are short and factual, providing a complementary, less forgiving stress test. We apply the identical sampling, inference, and scoring pipeline to both datasets.

For reproducible sampling, we draw question subsets using a fixed random seed (seed = 42) so that any researcher can reconstruct the exact evaluation set. To make this guarantee precise, we document the full deterministic data-loading and sampling pipeline: we load a version-pinned copy of the corpus (16,412 records; 14,984 distinct question strings), apply no question removal or preprocessing, and draw the subset with a single pandas `DataFrame.sample(` call over the loaded frame. Because the reproducibility of seeded sampling depends on the dataset version, row ordering, and the specific sampling implementation, we additionally release the exact sampled question set as Supplementary S1, regenerable with one command (`clinical-eval export-sample sample-random 50 sample-seed 42`). We evaluate $N{=}50$ randomly sampled questions *per dataset*, each subjected to 10 repeated inference runs per model, yielding 500 responses per model ~~(1,~~per dataset (2,500 ~~total~~per dataset; 5,000 in total across the five models and two datasets).

3.2 Models and Inference Setup

We select ~~three~~ four locally-deployable open-weight models ~~to span a range of parameter counts and~~ spanning 4B–12B parameters and two domain-specialization strategies, all served via the Ollama runtime [9], and include one closed-source API model as an external reference point:

- **Llama 3.1 8B** [3]: Meta AI's instruction-tuned general-purpose model. Represents the dominant open-weight baseline at the 8B scale.
- **Gemma 3 12B** [12]: Google DeepMind's instruction-tuned model at 12B parameters. Provides a larger general-purpose comparison point.
- **Gemma 3 4B** [12]: The 4B instruction-tuned sibling of Gemma 3 12B. It serves a dual role: a *scale* control against Gemma 3 12B and, crucially, a *same-size* general-purpose control against MedGemma 1.5 4B, which together isolate model scale from clinical fine-tuning.
- **MedGemma 1.5 4B** [8]: A Gemma-based 4B model fine-tuned on clinical corpora, testing the hypothesis that domain specialization compensates for smaller ~~scale.~~ scale—now tested directly against its same-size base model, Gemma 3 4B.
- **GPT-4.1-nano** (closed-source reference): OpenAI's smallest GPT-4.1-tier model, accessed via the vendor API. We include it solely as an external reference to contextualize the open models, *not* as a locally-deployable peer; it is therefore reported separately and excluded from "best open model" comparisons. Because closed APIs are versioned and may change without notice and their seed support is best-effort, we make no determinism claims for this model and run it only under the same decoding settings as the open models.

All ~~three~~ models receive an identical, fixed system instruction: *"You are a clinical expert. Answer in ≤6 sentences. Be direct. If uncertain, say so. Never recommend unsafe actions."* Holding the system prompt constant isolates model behavior from prompt-engineering effects, which is essential for a fair comparison.

Generation parameters are fixed across all models and all runs: temperature $T=0.2$, top-$p=1.0$, max tokens $= 512$, request timeout $= 120$ s with two retries on failure. The choice of $T=0.2$ is deliberate: this is a common production default intended to reduce variance. Our key research question is how much variance *remains* even in this near-deterministic regime.

3.3 Evaluation Metrics

We organize metrics into three groups.

Quality metrics (vs. medically validated ground-truth). Each generated response is compared to the gold answer using six complementary metrics: (1) **Exact Match**: binary equality after text normalization; (2) **Token F1**: token-level precision/recall F1 on lowercased alphanumeric tokens; (3) **String Similarity**: character-level SequenceMatcher ratio; (4) **BLEU** [10]: smoothed sentence-level n-gram overlap; (5) **ROUGE-L** [7]: longest common subsequence F1; (6) **BERTScore F1** [13]: contextual embedding similarity computed with `roberta-base`. Exact match and token F1 reward lexical fidelity; BERTScore captures semantic equivalence even when the phrasing differs.

LLM-as-judge score. A separate judge LLM evaluates each response against the gold answer using a structured rubric: *"Score from 0 to 1, where 1 means fully correct. Assess factual correctness and clinical safety."* The judge score provides a holistic clinical quality signal that complements lexical overlap metrics and is robust to valid paraphrases of the gold answer [14]. Concretely, we use a locally-hosted 20B-parameter open-source model served via Ollama as the judge. The judge inherits the same generation parameters as the target models ($T=0.2$, top-$p=1.0$, max tokens $= 512$) and is invoked once per response via the `/api/generate` endpoint (i.e., a single judge pass with no averaging across multiple runs). Scores are extracted from the judge output via regular-expression parsing with a fallback heuristic. Because the judge is itself stochastic~~and is not run multiple times, the reported judge scores carry unquantified variance; we discuss this limitation~~, the per-response judge scores in the main runs carry variance. To quantify the judge's own reliability, we additionally re-judge a model-stratified subset of responses five times and report the within-response score standard deviation and the one-way intraclass correlation coefficient (ICC(1), the standard test–retest reliability statistic) in Section 5.

Reproducibility metrics. For each (model, question) pair, we compute ~~two~~ the following metrics over the N repeated runs:

- **Normalized self-agreement rate** (lexical): fraction of runs whose normalized output matches the modal (most frequent) normalized output. A value of 1.0 indicates perfect consistency; $1/N$ indicates every run produced a distinct response.
- **Normalized response uniqueness rate** (lexical): number of distinct normalized outputs divided by N. A value of 1.0 indicates every run was unique.
- **Semantic self-similarity**: the mean pairwise semantic similarity, measured as BERTScore-F1 [13], across the N run outputs for a given (model, question) pair. Unlike the two lexical measures, this credits runs that are differently worded but semantically equivalent, directly addressing the concern that paraphrase inflates apparent run-to-run variability. We caution that embedding-based similarity has a high baseline floor, so it should be read as a complement to—not a replacement for—the lexical measures.

~~Both~~ The two lexical metrics operate on text normalized by lowercasing, whitespace collapsing, and removal of non-alphanumeric characters, making them robust to trivial formatting differences~~. These two~~; the semantic metric operates on the raw generated text. Together these metrics are complementary: self-agreement emphasizes the *stability* of the most likely output, ~~while~~ uniqueness measures the *breadth* of the output distribution, and semantic self-similarity isolates *meaning-level* agreement from surface phrasing.

Efficiency. We record per-run latency (ms) and throughput (tokens/second) from the inference server, as practical deployment constraints frequently make efficiency a decisive factor.

3.4 Pipeline Architecture

The pipeline is implemented as a Python package with a CLI exposing three subcommands:

- `clinical-eval run`: queries each model for each question N times and stores raw responses as JSONL/CSV/Parquet.
- `clinical-eval score`: applies the full metric suite to every stored response.
- `clinical-eval report`: aggregates scored responses and renders the ~~six-section~~ Markdown report.

The separation of generation from scoring is a key design choice: it allows the scoring step (including new metrics) to be re-run without re-querying the LLMs, which is expensive and introduces additional variance into the experiment. The pipeline is fully resumable: interrupted runs continue from where they left off: making it practical on commodity hardware where long jobs may be preempted. All raw outputs are retained as an audit trail, enabling downstream re-analysis without re-running models. Inference is fully stateless across questions and runs: each query is an independent, single-turn generation request with no conversation history carried over between queries, so there is no cross-sample context leakage that could otherwise inflate or deflate measured reproducibility.

A shared system prompt, fixed random seed, and version-pinned dependencies ensure that any researcher can reproduce the exact experimental conditions by cloning the repository and running `clinical-eval all config-pathconfigs/pipeline.yaml`.

4 Results

Tables 1 and 2 summarize results across ~~1,500 responses~~ both datasets (50 questions × 10 runs × ~~3 models)~~5 models per dataset; 2,500 responses per dataset, 5,000 in total), and Figs 1 and 2 visualize the same model-level estimates with their bootstrap confidence intervals.

Aggregation and uncertainty. We state the aggregation procedure explicitly. For the quality metrics, a model's score is the mean over all of its responses; because the design is balanced (10 runs per question), this equals the mean across questions of the per-question mean. For the reproducibility metrics, we compute each metric per (model, question) and then average over questions. To avoid over-interpreting small between-model differences, every model-level value is reported with a 95% confidence interval obtained by a clustered percentile bootstrap that resamples questions with replacement (1,000 resamples); per-question standard deviations are additionally available in the released aggregates.

Table 1: Quality metrics: model output vs. gold answer, mean [95% CI]. Per dataset, n=50 questions × 10 runs; each value is the across-question mean with a clustered percentile-bootstrap 95% CI (1,000 resamples). Bold = best among the open-weight models per column, per dataset. †GPT-4.1-nano is a closed-source API reference, not a peer of the open models, and is excluded from the "best" comparison.

Model	BERTScore F1	Token F1	ROUGE-L	Judge
MedQuAD				
Llama 3.1 8B	**0.848** [0.837, 0.857]	**0.287** [0.264, 0.311]	**0.172** [0.154, 0.189]	0.565 [0.480, 0.643]
Gemma 3 12B	0.843 [0.833, 0.852]	0.238 [0.218, 0.258]	0.147 [0.133, 0.160]	**0.580** [0.477, 0.669]
Gemma 3 4B	0.845 [0.836, 0.854]	0.245 [0.225, 0.267]	0.151 [0.137, 0.166]	0.460 [0.371, 0.554]
MedGemma 1.5 4B	0.846 [0.835, 0.855]	0.244 [0.224, 0.263]	0.155 [0.140, 0.172]	0.452 [0.369, 0.545]
GPT-4.1-nano†	0.848 [0.838, 0.859]	0.266 [0.241, 0.291]	0.165 [0.146, 0.182]	0.733 [0.656, 0.798]
MedicationQA				
Llama 3.1 8B	0.849 [0.841, 0.855]	**0.209** [0.184, 0.233]	**0.134** [0.118, 0.149]	0.600 [0.507, 0.696]
Gemma 3 12B	0.848 [0.841, 0.855]	0.185 [0.160, 0.209]	0.121 [0.106, 0.137]	**0.735** [0.646, 0.816]
Gemma 3 4B	0.842 [0.834, 0.848]	0.175 [0.148, 0.202]	0.110 [0.095, 0.123]	0.680 [0.573, 0.781]
MedGemma 1.5 4B	**0.851** [0.843, 0.858]	0.200 [0.174, 0.226]	0.127 [0.112, 0.143]	0.640 [0.539, 0.731]
GPT-4.1-nano†	0.852 [0.844, 0.859]	0.209 [0.182, 0.238]	0.133 [0.116, 0.150]	0.760 [0.662, 0.847]

Table 2: Within-model reproducibility across 10 runs, mean [95% CI]. Self-agreement (higher is better; $1/N=0.10$ = every run distinct) and uniqueness (lower is better) are lexical; semantic self-similarity is the mean pairwise BERTScore-F1 across runs (higher is better, robust to paraphrase). Bold = best among open-weight models per column, per dataset; †closed-source reference (excluded from "best").

Model	Self-agreement↑	Uniqueness↓	Semantic self-sim.↑
MedQuAD			
Llama 3.1 8B	0.106 [0.100, 0.116]	0.994 [0.984, 1.000]	0.937 [0.932, 0.941]
Gemma 3 12B	**0.190** [0.158, 0.230]	**0.896** [0.848, 0.934]	**0.959** [0.954, 0.964]
Gemma 3 4B	0.162 [0.142, 0.184]	0.926 [0.898, 0.950]	0.958 [0.953, 0.962]
MedGemma 1.5 4B	0.138 [0.118, 0.164]	0.952 [0.922, 0.978]	0.947 [0.942, 0.952]
GPT-4.1-nano†	0.150 [0.128, 0.176]	0.944 [0.914, 0.970]	0.958 [0.953, 0.962]
MedicationQA			
Llama 3.1 8B	0.162 [0.126, 0.208]	0.930 [0.882, 0.972]	0.939 [0.934, 0.944]
Gemma 3 12B	**0.260** [0.222, 0.302]	**0.764** [0.700, 0.820]	**0.965** [0.959, 0.970]
Gemma 3 4B	0.176 [0.152, 0.204]	0.892 [0.848, 0.928]	**0.965** [0.960, 0.969]
MedGemma 1.5 4B	0.200 [0.160, 0.246]	0.868 [0.814, 0.916]	0.952 [0.946, 0.957]
GPT-4.1-nano†	0.190 [0.156, 0.226]	0.884 [0.826, 0.932]	0.959 [0.955, 0.964]

Table 3: Inference efficiency (MedQuAD, tokens/second). Median per-run throughput; MedicationQA is comparable. GPT-4.1-nano† is API-served and network-bound rather than compute-bound on local hardware.

Model	Tokens/s
GPT-4.1-nano†	97.9
Gemma 3 4B	61.0
Llama 3.1 8B	41.3
MedGemma 1.5 4B	29.1
Gemma 3 12B	23.7

Quality. Across both datasets, BERTScore F1 is tightly clustered (~~0.847~~0.84–0~~.852) , confirming broadly similar *semantic* alignment with gold at the embedding level. Differences become clearer on lexical metrics:~~ .85) for all five models, indicating broadly similar embedding-level alignment with the gold answers and limited discriminative power at the semantic-similarity level. The lexical and judge metrics separate the models more clearly. On MedQuAD, Llama 3.1 8B leads the open models on Token F1 (~~0.277 vs. 0.239–0.249) and BERTScore (0.852). The LLM judge—which rewards holistic clinical correctness—tells a subtly different story:~~ 0.287) and ROUGE-L (0.172), whereas Gemma 3 12B ~~edges ahead (0.600)over Llama~~obtains the highest open-model judge score (0.580); on MedicationQA the judge scores are higher overall and Gemma ~~3.1~~3 ~~8B (0.592), while the clinically fine-tuned MedGemma 1.5 4B lags substantially (0.459)~~12B again leads the open models (0.735). The closed-source reference, GPT-4.1-nano, obtains the highest judge score on both datasets (0.733 and 0.760), but its BERTScore, Token F1, and ROUGE-L all fall within the range spanned by the open models, so it does not dominate the comparison—a useful calibration point rather than an upper bound the open models fail to approach. Notably, no model produces an exact lexical match with any gold answer (exact match = 0.000 across all models), confirming that all ~~three~~ models

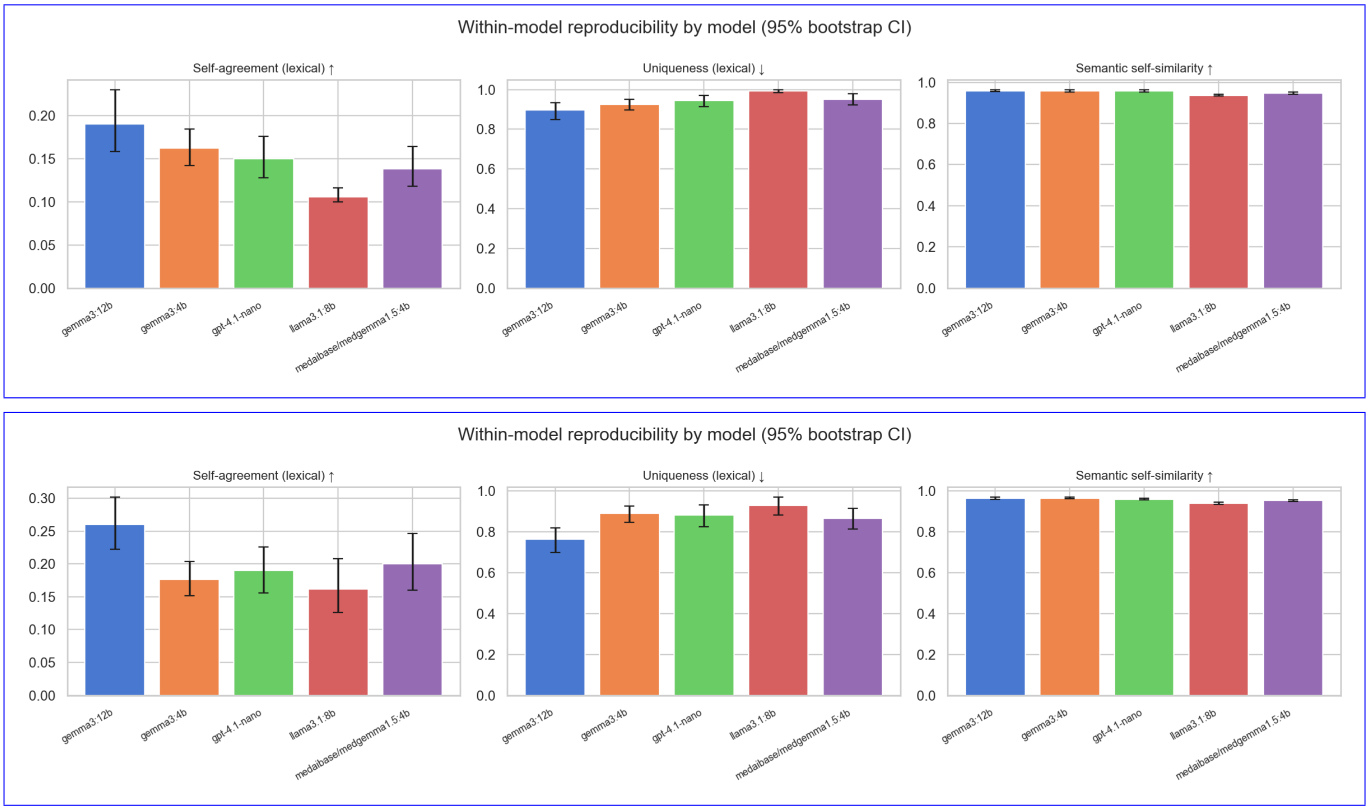

Figure 1: Within-model reproducibility by model, with 95% bootstrap confidence intervals. Per-model lexical self-agreement (↑), lexical uniqueness (↓), and semantic self-similarity (↑) across the 10 runs. (A) MedQuAD (top); (B) MedicationQA (bottom). Across both datasets and all five models, lexical agreement is low while semantic self-similarity is high (0.94–0.97), showing that run-to-run divergence is largely paraphrastic rather than meaning-level.

paraphrase rather than recall~~. Nevertheless, this was important to show also~~ ; we also read this as a signal of no overfitting ~~of the models for the dataset we used as~~ to the medically validated ground-truth corpus.

Reproducibility. The ~~most striking finding is the near-complete output uniqueness at T=0.2. Out of 500 runs per model, Gemma 3 12B produced 434 unique normalized responses (86.8%), MedGemma 1.5 4B produced 468 (93.6%),~~ central finding is a large, consistent gap between *lexical* and ~~Llama 3.1 8B produced 487~~ *semantic* reproducibility that holds across both datasets and all five models, including the closed reference (Fig 1). Lexically, outputs are highly variable: self-agreement ranges from 0.11 to 0.26 and uniqueness from 0.76 to 0.99, so even the most consistent model reproduces its modal answer on only a minority of runs (~~97.4%). Self-agreement rates, the fraction of runs matching the modal response—range from 0.122 (~~e.g., on MedQuAD, Llama 3.1 8B ~~) to 0.198 (~~is unique on 99.4% of runs and Gemma 3 12B ~~): even the most consistent model agrees with itself on fewer than one in five runs on average. Pairwise~~ on 89.6%). Semantically, however, the same runs are highly similar: mean pairwise semantic self-similarity is 0.94–0.97 for every model on both datasets. In other words, run-to-run divergence at

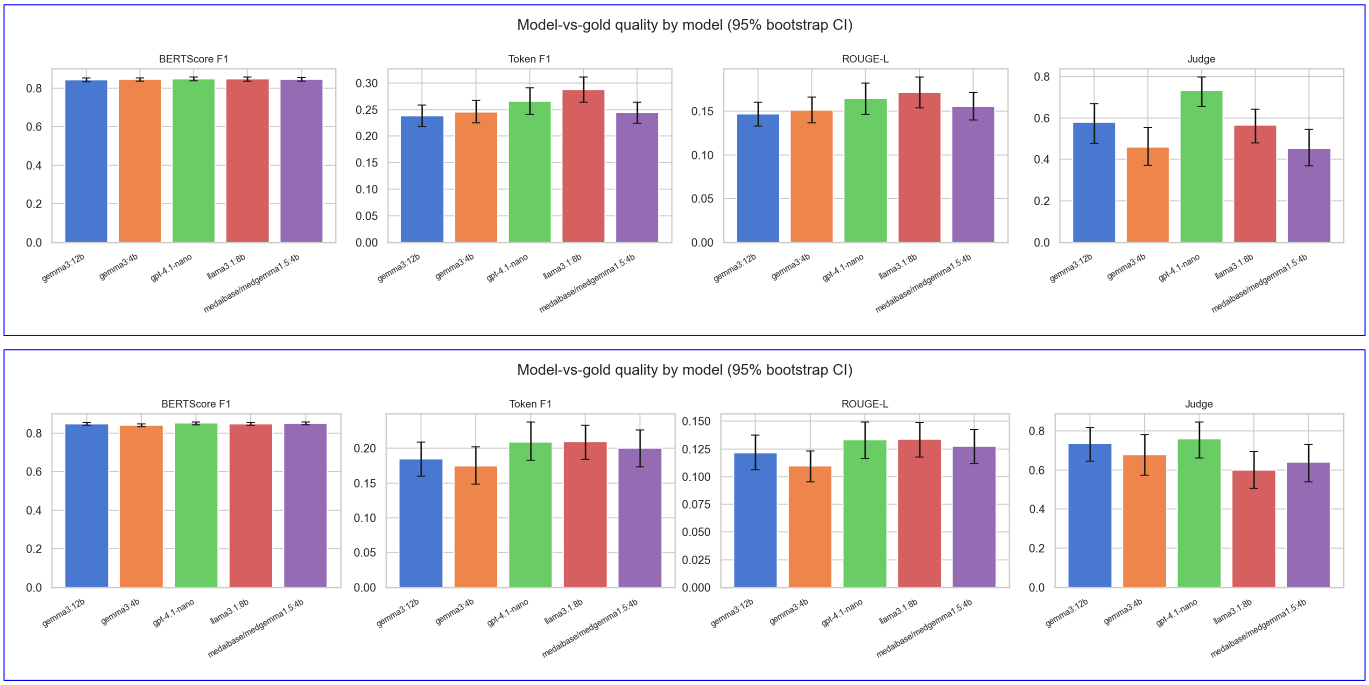

Figure 2: Model-vs-gold quality by model, with 95% bootstrap confidence intervals. BERTScore F1, Token F1, ROUGE-L, and LLM-judge score per model. (A) MedQuAD (top); (B) MedicationQA (bottom). BERTScore is tightly clustered across models; the closed reference GPT-4.1-nano leads on the judge score but remains within the open models' range on the other metrics.

T=0.2 is overwhelmingly *paraphrastic* rather than meaning-level—models say substantially the same thing in different words. This directly addresses the concern that purely lexical metrics overstate instability by counting paraphrases as disagreements; we nonetheless read the semantic figure as a complement to, not a replacement for, the lexical measures, given the known high baseline floor of embedding similarity. Pairwise *exact* output overlap across ~~all~~ model pairs is ~~exactly~~ 0.000, ~~confirming that the three models occupy entirely disjoint output spaces.~~ which we treat only as a qualitative observation: strict string comparison is uninformative about semantic or clinical equivalence—two medically equivalent but differently worded answers count as non-overlapping (Section 3.3). We therefore frame these results as run-to-run *output variability*, not as direct evidence of clinical inconsistency.

Quality–reproducibility trade-off and the role of scale. A trade-off is ~~apparent across the three~~ again visible among the open models. Llama 3.1 8B ~~achieves~~ attains the highest lexical quality and ~~fastest throughput (43.0 tok/s) but the worst reproducibility~~throughput on MedQuAD but is the least lexically reproducible (uniqueness 0.99) and has the lowest semantic self-similarity. Gemma 3 12B ~~achieves the highest judge score and best reproducibility~~is the most reproducible model on both datasets—on both the lexical and the semantic measures—while also achieving the best open-model judge score, at the cost of ~~being the slowest (25.5 tok/s).~~ the lowest throughput (Table 3). With the same-size control now included, the effect of scale is clean: Gemma 3 12B is more reproducible and higher-scoring than Gemma 3 4B, whereas the clinically fine-tuned MedGemma 1.5 4B ~~ranks last or second-to-last on every metric despite its clinical fine-tuning. However, because MedGemma is also the smallest model in our comparison (4B vs. 8B and 12B), this result conflates the effect of domain adaptation with a 2–3× parameter count difference; a controlled comparison against a~~ does *not* improve over its same-size general-purpose ~~baseline (e.g.,~~ sibling Gemma 3 4B ~~) would be needed to disentangle these factors~~(judge 0.452 vs. 0.460 on MedQuAD; 0.640 vs. 0.680 on MedicationQA, with overlapping confidence intervals), indicating that at these scales capacity, not domain fine-tuning, drives the differences (see Section 5).

5 Discussion and Caveats

Reproducibility is not solved by low temperature. The most actionable finding is that $T{=}0.2$ does not yield *lexically* reproducible behavior in practice. Self-agreement rates of ~~0.12~~0.11–0~~.20~~.26 mean that for a given clinical question, the same model ~~is almost certain to give a different answer~~ almost always returns differently worded text on the next invocation: Llama 3.1 8B, the least consistent model, produces a unique normalized response on ~~97.4~~up to 99% of runs. ~~For clinical deployment, this necessitates~~ Importantly, this instability is largely surface-level: semantic self-similarity across runs stays high (0.94–0.97), so the dominant risk is variable phrasing and emphasis rather than outright contradictory clinical content. Even so, variable phrasing can shift perceived meaning, complicate auditing, and undermine user trust, so for clinical deployment we still recommend output-stabilization strategies: majority voting across multiple runs, confidence-score gating, or mandatory human-in-the-loop review before any LLM output is surfaced to a clinician. Single-pass benchmarks that report average accuracy without also reporting within-model variance conceal this risk entirely. Here we would also like to emphasize that we did not use $T{=}0.0$ not only because it is not common for question answering GenAI tasks, but also as a gatekeeper of training bias in case the dataset of medically validated ground-truth or portions of it was visible during the model(s) training.

Clinical fine-tuning versus model scale. ~~MedGemma~~Including Gemma ~~1.5~~3 4B ~~'s clinical fine-tuning did not translate to higher judge scores, better lexical quality, or better reproducibility relative to the larger~~ as a same-size general-purpose ~~models. On every metric in Tables 1 and 2, MedGemma finishes last or second-to-last. Critically, however, this comparison is confounded by model size: MedGemma (4B) is 2–3× smaller than Llama~~control lets us separate domain fine-tuning from model scale—the comparison the original study could only flag as a confound. Two patterns emerge. First, *scale helps*: Gemma ~~3.1~~3 ~~8B and~~ 12B outperforms Gemma 3 ~~12B, so the observed gap may reflect a capacity difference rather than a failure of domain adaptation. Isolating the fine-tuning effect would~~

~~require comparing MedGemmaagainst a~~ 4B on judge score and is more reproducible on both datasets. Second, at fixed size, *clinical fine-tuning does not*: MedGemma 1.5 4B matches or trails its same-size ~~general-purpose baseline (e.g.,~~ base model Gemma 3 4B ~~), which we leave to future work. This finding~~ on the judge score (0.452 vs. 0.460 on MedQuAD; 0.640 vs. 0.680 on MedicationQA) and shows no reproducibility advantage, with confidence intervals that overlap throughout. The earlier apparent "MedGemma underperforms" effect was therefore largely a capacity effect: once size is held constant, domain adaptation confers no measurable benefit on these consumer-health QA tasks. This is consistent with the broader observation that scale often dominates domain adaptation when the target task is well covered in pretraining [11, 18]. Practitioners should evaluate models empirically rather than relying on domain-specific branding, and should prefer same-scale comparisons to isolate the contribution of fine-tuning.

LLM-as-judge introduces its own variance. The judge model (a 20B open-source model, $T{=}0.2$) provides a clinically meaningful holistic quality signal, but is itself stochastic~~. In this study,~~ , and in the main runs each response was judged ~~exactly once; we did not run multiple judge passes or average across them.~~ once. To characterize this, we re-judged a model-stratified subset of responses five times on each dataset. The judge was highly stable on MedQuAD—mean within-response score standard deviation 0.064 and one-way intraclass correlation ICC(1) = 0.94 ($n{=}22$ responses)—and moderately stable on MedicationQA (within-response SD 0.111, ICC(1) = 0.69, $n{=}25$), indicating that most of the score variance on MedQuAD, and a substantial part on the harder MedicationQA items, reflects genuine between-response differences rather than judge noise. We separately note that the judge returned an unparseable score on a small fraction of responses (5.2% on MedQuAD, 2.9% on MedicationQA); these were excluded from the judge means, and all other metrics are unaffected. Because the judge score is also the metric on which models diverge most~~(0.459 vs. 0.600), this unquantified judge variance is~~ , this residual variance remains a meaningful limitation. For production evaluations, we recommend running the judge multiple times per response and averaging, or adopting a deterministic rubric-grading approach where feasible.

5.1 Toward improved reproducibility: type-3 ANFIS and related methods

Our metrics *diagnose* reproducibility; a natural next question is how to *improve* it. Several complementary directions exist, and we position our contribution relative to them. **Self-consistency** decoding [15, 16] samples multiple generations and returns the most frequent answer; it is the output-stabilization analogue of the variability we measure, and our repeated-inference protocol provides exactly the samples such a scheme would aggregate. **Semantic entropy** [19] clusters generations by meaning and estimates uncertainty in semantic space to flag confabulations;

our semantic self-similarity metric is a lightweight, deployment-oriented cousin of this idea, trading a full entropy estimate for an interpretable pairwise-agreement score. **Conformal prediction** [20] offers distribution-free guarantees and could wrap a model's repeated outputs in calibrated prediction sets, converting raw variability into a formal coverage statement suitable for safety-critical review.

A further promising direction, and one we highlight at the editor's suggestion, is to model the higher-order uncertainty we observe with a **type-3 fuzzy logic / ANFIS** layer. Type-3 fuzzy systems extend type-1 and interval type-2 systems with an additional level of membership-function uncertainty, which makes them well suited to representing the "uncertainty about uncertainty" that arises when a model returns many semantically related but non-identical answers across runs [21, 22]. Concretely, an adaptive neuro-fuzzy inference system (ANFIS) with type-3 antecedents could act as an uncertainty-aware aggregator over the N repeated generations, mapping features of the run set (e.g., pairwise semantic similarity, judge-score spread, response-length dispersion) to a calibrated stability/confidence score that gates whether an answer is surfaced, escalated, or withheld. We leave a full implementation and evaluation of such a type-3 ANFIS post-processing layer to future work, but note it as a principled route from *measuring* reproducibility to *actively managing* it in deployment.

Inference environment. All results were obtained via the Ollama local inference server on a single workstation. Quantization level, hardware configuration, and batch size can affect generation quality and throughput; API-hosted counterparts of the same model families may exhibit different variance profiles.

6 Conclusion

We have presented and applied a reproducibility-first evaluation framework for small open-weight LLMs in clinical question answering, covering ~~1,500 responses from three models evaluated on 50 MedQuAD questions~~ 5,000 responses from five models (four open-weight, plus a closed-source reference) evaluated on two consumer-health datasets (MedQuAD and MedicationQA), 50 questions each with 10 runs~~each~~. The results lead to three concrete takeaways for practitioners.

First, **low temperature does not buy ~~content~~ lexical reproducibility, but the variability is mostly paraphrastic**. Even at T=0.2, lexical self-agreement ~~rates hover between 0.12 and 0.20, and 87–97% of all~~ is only 0.11–0.26 and 76–99% of outputs per model are unique~~. Any medical~~; yet semantic self-similarity stays at 0.94–0.97, so models largely restate the same content in different words. Any medically related usage must still account for this surface variance through ensembling, gating, or human review.

Second, **at these parameter counts, model scale—not clinical ~~fine-tuning does not compensate for a large capacity gap~~fine-tuning—drives quality**. With a same-size control (Gemma 3 ~~12B (general-purpose, 12B) outperforms~~ 4B) now included, MedGemma 1.5 4B ~~(clinically fine-tuned, 4B) on every dimension: judge score, lexical quality, and reproducibility. However, this comparison confounds fine-tuning with a 3× scale difference; a same-size comparison (e.g.,~~ shows no advantage over its general-purpose sibling on either dataset, while the larger Gemma 3 ~~4B vs. MedGemma 1.5 4B) is needed to isolate the fine-tuning~~ 12B leads the open models. The earlier apparent shortfall of the clinically fine-tuned model was largely a capacity effect. Model selection should be empirically driven, not label-driven.

Third, **quality and reproducibility partially trade off**. Llama 3.1 8B maximizes lexical quality and speed but minimizes consistency; Gemma 3 12B offers the best balance of quality and reproducibility at the cost of throughput. The right choice depends on whether the deployment context prioritizes answer fidelity, consistency, or latency.

Our open-source pipeline makes this three-dimensional evaluation accessible to any team with a commodity workstation, and we invite the clinical NLP community to adopt reproducibility metrics as a standard alongside accuracy in LLM benchmarking.

Future work. Planned extensions include additional models (Phi-3, Mistral-Medical, Qwen-Med), ~~a controlled comparison of same-size general-purpose and clinically fine-tuned models~~ evaluation on further benchmarks (e.g., ~~Gemma 3 4B vs. MedGemma 1.5 4B)to isolate the effect of domain adaptation from model scale, evaluation on USMLE and MedQA benchmarks, ensemble decoding strategies to~~ USMLE-style MedQA), ensemble and type-3 ANFIS decoding/aggregation strategies to actively improve reproducibility, ~~and a calibration study~~ a human spot-check of the LLM judge against expert clinician ratings, and evaluation on a real-world EHR-derived question set (e.g., MIMIC-IV [6]).

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